

eSpark Consulting Group

Project Proposal

Comprehensive Proposal for PCB Design and Development

Project Overview

This proposal outlines the process for designing and delivering a high-quality PCB, addressing all requirements for functionality, manufacturability, and performance. The project involves reverse engineering, replacing obsolete parts, designing schematics, creating a components library, PCB layout, and thorough testing. Our focus is on adherence to industry standards and delivering all necessary files for successful production.

Milestones with Key Features and Quotations

M1: Research and Analysis

- **Price: $640**
- **Timeline:** Week 1
- **Key Features:**
  - Reverse engineering of existing designs.
  - Identification and replacement of obsolete components.
  - Comprehensive analysis for improving efficiency and compatibility.

M2: Schematic and Components Library Design

- **Price: $640**
- **Timeline:** Week 2
- **Key Features:**
  - Development of detailed schematics aligned with requirements.
  - Creation and validation of a complete components library, ensuring availability and compatibility.

M3: PCB Design and Routing

- **Price: $960**
- **Timeline:** Weeks 3-4
- **Key Features:**
  - PCB redesign with adherence to industry standards.
  - Integration of all OEM-required information.
  - Routing optimization for performance and manufacturability.

M4: Review and Preparing Deliverables for OEM

- **Price: $320**
- **Timeline:** Week 5
- **Key Features:**
  - Delivery of complete Gerber files, BOM, and CPL.
  - Comprehensive design review and documentation.
  - Support for addressing OEM queries and implementing required changes.

M5: Final Testing and Validation

- **Price: $640**
- **Timeline:** Week 6
- **Key Features:**
  - Functional and performance testing of the PCB.
  - Validation against design specifications and operational standards.

Deliverables

The following materials will be delivered upon project completion:

1. **Gerber Files**: For PCB fabrication.
2. **Bill of Materials (BOM)**: Complete component listing with specifications.
3. **Pick-and-Place File (CPL)**: For automated assembly.
4. **Schematic Diagrams**: For reference and troubleshooting.
5. **Design Review Document**: Highlights key design choices and compliance.
6. **Testing Report**: Includes results from functional and validation testing.
7. **OEM Support Documents**: Responses and solutions for any required changes during PCBA.

Timeline and Pricing

- **Total Price: $3,200**
- **Total Timeline:** 5 to 6 weeks **(excluding fabrication and assembly time)**.

This timeline includes the following key activities:

- Complete reverse engineering.
- Research and analysis to replace obsolete/outdated parts.
- Redesign of PCB with proper standards and all OEM-required information.
- Support in PCBA (answering OEM queries and implementing change requirements).

Conclusion

This proposal outlines a clear and detailed roadmap for delivering a comprehensive PCB design solution that meets your needs. By breaking the project into structured milestones, we ensure transparency, quality, and timely delivery. With a focus on adherence to industry standards and robust support during PCBA, we aim to provide a design that is both reliable and manufacturable.

We are committed to collaborating closely with you throughout the process to address any queries and ensure the final product exceeds expectations.

We look forward to partnering with you on this project and delivering exceptional results.